AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Experiment Title:	Study of Analog to Digital Conversion using MATLAB		
Experiment No:	04	Date of Submission:	3 February 2026
Course Title:	**DATA COMMUNICATION**		
Course Code:	**01936**	Section:	D
Semester:	Spring 2025-26	Course Teacher:	MOHAMMAD ASADUZZAMAN KHAN

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Group Name/No.: 07

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FACULTY COMMENTS		
	Marks Obtained	
	Total Marks	

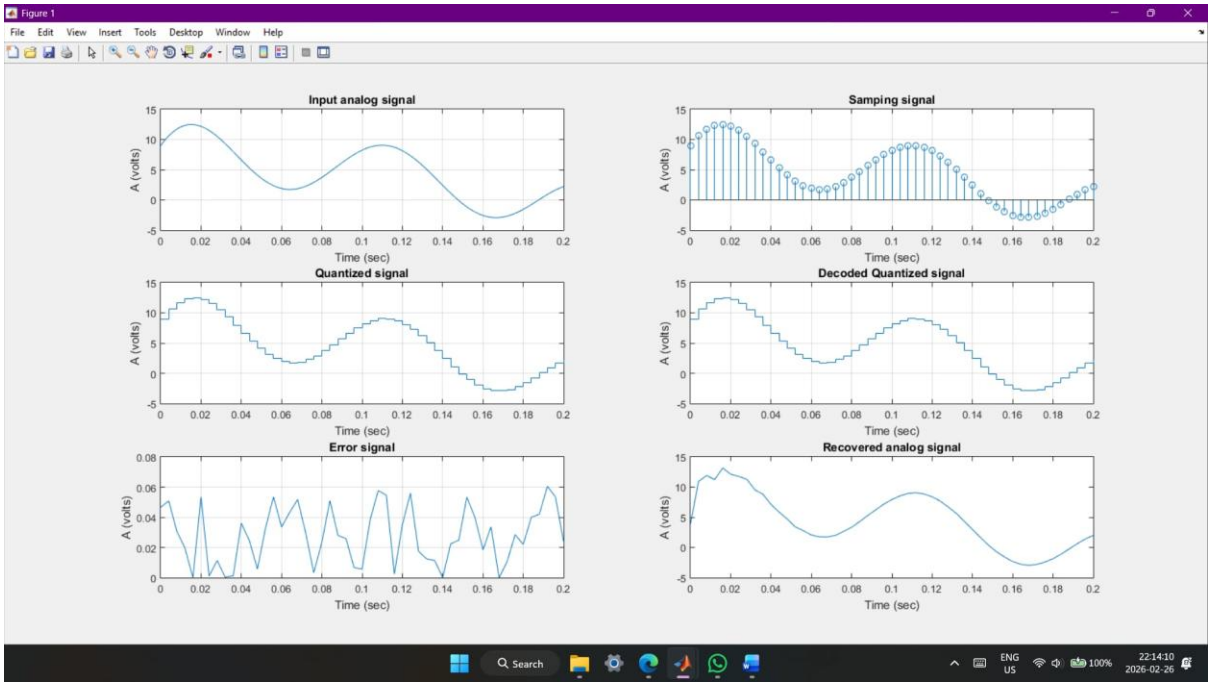
Code:

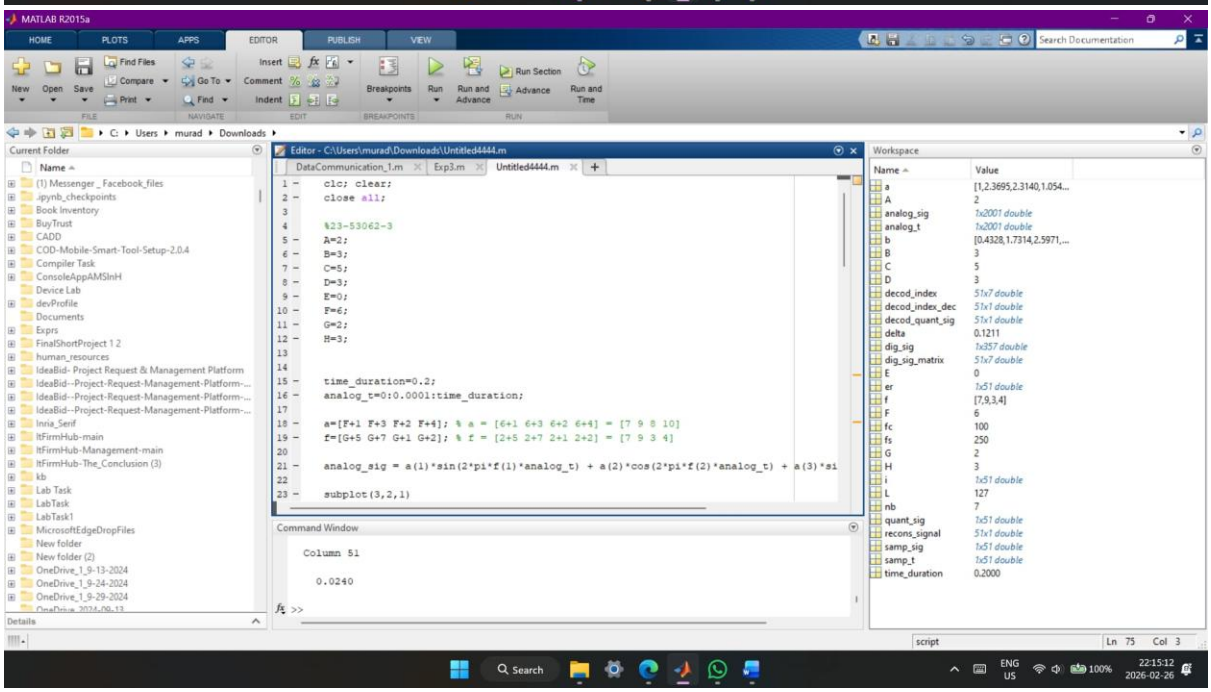
```
clc; clear;
close all;

%ID = AB-CDEFG-H = 23-53062-3
A=2;
B=3;
C=5;
D=3;
E=0;
F=6;
G=2;
H=3;

time_duration=0.2;
analog_t=0:0.0001:time_duration;

a=[F+1 F+3 F+2 F+4];    % a = [6+1 6+3 6+2 6+4] = [7 9 8 10]
f=[G+5 G+7 G+1 G+2];   % f = [2+5 2+7 2+1 2+2] = [7 9 3 4]

%Analog signal:
analog_sig = a(1)*sin(2*pi*f(1)*analog_t) + a(2)*cos(2*pi*f(2)*analog_t) +
a(3)*sin(2*pi*f(3)*analog_t);

subplot(3,2,1)
plot(analog_t,analog_sig)
xlabel('Time (sec)')
ylabel ('A (volts)')
title ('Input analog signal')
grid on;

% % sampling
fs=250;
samp_t=0:1/fs:time_duration;
samp_sig = a(1)*sin(2*pi*f(1)*samp_t) + a(2)*cos(2*pi*f(2)*samp_t) +
a(3)*sin(2*pi*f(3)*samp_t);

subplot (3,2,2)
stem(samp_t,samp_sig)
xlabel('Time (sec)')
ylabel ('A (volts)')
title ('Samping signal')
grid on;
%
```

```
% % quantization
nb=7;
L=2^nb-1;
delta=(max(samp_sig)-min(samp_sig))/L;
i= round ((samp_sig-min(samp_sig))/delta);
quant_sig=min(samp_sig)+i*delta;

subplot (3,2,3)
stairs(samp_t,quant_sig)
xlabel('Time (sec)')
ylabel ('A (volts)')
title ('Quantized signal')
grid on;
%
% % encoding
dig_sig_matrix=de2bi(i,nb);
% encoded digigal signal
dig_sig=reshape (dig_sig_matrix',1,[])
%
% % decoding
decod_index=reshape(dig_sig,nb,[])';
decod_index_dec=bi2de(decod_index);
decod_quant_sig=min(samp_sig)+decod_index_dec*delta;
%
subplot (3,2,4)
stairs(samp_t,decod_quant_sig)
xlabel('Time (sec)')
ylabel ('A (volts)')
title ('Decoded Quantized signal')
grid on;
%
% % error signal
er=abs(samp_sig-quant_sig)
%
subplot(3,2,5)
plot(samp_t,er)
xlabel('Time (sec)')
ylabel ('A (volts)')
title ('Error signal')
grid on;
%
% % Low pass filter
fc=100;
[b,a]=butter(4,fc/(fs/2)); % Low passfilter
recons_signal=filter(b,a,decod_quant_sig);
%
subplot(3,2,6)
```

```
plot(samp_t,recons_signal)
xlabel ('Time (sec)')
ylabel ('A (volts)')
title ('Recovered analog signal')
```

Output:





```
dig_sig =

  Columns 1 through 18

    0    1    0    0    0    1    1    0    0    0    0    1    1    1    1    0    0    1

  Columns 19 through 36

    1    1    1    0    1    1    1    1    1    1    1    1    1    1    1    1    1    1

  Columns 37 through 54

    0    1    1    1    1    1    1    1    1    0    1    1    1    1    1    1    1    0

  Columns 55 through 72

    1    1    1    0    1    0    0    1    1    0    1    0    1    1    0    1    1    1

  Columns 73 through 90

    1    1    0    0    1    0    0    1    0    0    0    1    0    1    0    1    1    1

  Columns 91 through 108

    0    0    1    0    0    1    1    0    0    0    1    1    0    1    0    0    0    0

  Columns 109 through 126

    1    0    1    0    0    1    1    0    0    1    0    1    1    1    0    0    1    0

  Columns 127 through 144

    1    1    0    1    0    1    0    0    0    0    0    1    1    0    1    1    1    0

  Columns 145 through 162

    1    1    0    1    1    1    1    1    1    0    1    1    1    0    0    0    1    1

  Columns 163 through 180

    1    1    1    0    0    1    0    1    1    0    1    0    1    0    0    1    1    1

  Columns 181 through 198

    0    1    0    0    0    0    0    1    1    1    1    0    0    0    1    1    0    1

  Columns 199 through 216

    0    0    0    1    1    0    0    0    0    0    1    1    1    1    0    1    1    0

  Columns 217 through 234

    1    0    0    1    0    1    0    1    0    0    1    1    0    0    1    0    1    0
```

Columns 235 through 252

0 0 0 1 1 1 1 0 1 1 0 0 0 1 1 0 1 0

Columns 253 through 270

1 0 0 0 0 1 0 1 1 1 0 1 0 0 1 1 1 1

Columns 271 through 288

0 0 0 0 0 0 1 0 0 0 1 1 0 0 0 0 0 0

Columns 289 through 306

0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0

Columns 307 through 324

0 0 0 1 1 0 0 0 0 1 1 0 1 0 0 0 0 1

Columns 325 through 342

0 0 1 0 0 1 0 0 1 1 0 0 0 0 0 0 0 1

Columns 343 through 357

0 0 1 1 0 0 1 0 0 1 0 1 0 1 0

er =

Columns 1 through 10

0.0467 0.0508 0.0307 0.0196 0.0000 0.0536 0.0010 0.0114 0.0003 0.0013

Columns 11 through 20

0.0361 0.0246 0.0056 0.0327 0.0535 0.0335 0.0432 0.0518 0.0294 0.0033

Columns 21 through 30

0.0233 0.0510 0.0279 0.0258 0.0067 0.0057 0.0384 0.0577 0.0546 0.0024

Columns 31 through 40

0.0352 0.0560 0.0175 0.0124 0.0114 0.0001 0.0224 0.0248 0.0533 0.0399

Columns 41 through 50

0.0183 0.0336 0 0.0107 0.0285 0.0220 0.0400 0.0420 0.0605 0.0534

Column 51

0.0240